MIL-B-52796A(ME)
23 December 1977
SUPERSEDING
MIL-B-52796(ME)
11 April 1974

MILITARY SPECIFICATION

BOLTS, RIBBED NECK, STRUCTURAL TYPE, WITH RECESSED LOCKNUT

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers button-head and countersunk head, threaded, interrupted rib, interference body type, and structural type bolts with heavy hex nuts, recessed hex locknuts, and hardened structural washers.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage of.

FSC 5306

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specification and standard required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request or proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A325 - High Strength Bolts for Structural Joints, Including Suitable Nuts and Plain Hardened Washers.
F436 - Hardened Washers for use with Structure Bolts.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Threads Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington DC 20402.)

3. REQUIREMENTS

3.1 Description. The bolts and nuts shall be as shown in Tables I and II, Figure 1 and as specified herein. The ribs shall be tapered for ease in driving. The nuts shall be hexagon, recessed, and self locking. When specified, washers shall be furnished (see 6.2 and 6.4).

3.2 Material. Bolts and nuts shall be carbon manganese steel conforming to ASTM A325, Type 1, and shall have a minimum tensile strength of 120,000 pounds per square inch (psi). Washers furnished as a part of the bolt shall be of hardened steel in accordance with ASTM F436 and as in Table II (see 6.5).

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3.3 Fabrication.

3.3.1 Threads. Threads shall be of the UNC series and as specified in Table I, Figure 1, and NBS Handbook H28.

3.3.2 Boltheads. Boltheads shall be button head or countersunk head as specified (see 6.2), in accordance with Tables I and II, and Figure 1.

3.3.3 Size. The bolt type, diameter, length, thread length, and number of ribs shall be in accordance with Table I, Figure 1, and as specified herein.

3.3.3.1 Commercial sizes. Bolts of standard commercial sizes not covered in Table I shall be in accordance with ASTM A325 and Table II (see 6.2).

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Table I. Bolt Type and Dimensions

NSN	Bolt Length L	Rib Length R	Thread Length T	UNC Type Threads	Major Diameter of Ribs(C)	Number of Ribs	Type of Head	Diameter of Head A	Height of Head H
5306-00-240-0802	1.500	.469	1.031	3/4-10	0.828	17	Button Head	1.375	.469
5306-00-240-0803	1.655	0.594	1.031	3/4-10	0.828	17	Button Head	1.375	.469
5306-00-240-0804	1.875	0.844	1.031	3/4-10	0.828	17	Button Head	1.375	.469
5306-00-240-0805	2.062	1.031	1.031	3/4-10	0.828	17	Button Head	1.312	.562
5306-00-240-0806	2.625	1.031	1.031	3/4-10	0.828	17	Button Head	1.312	.562
5306-00-240-0807	2.438	1.031	1.031	3/4-10	0.828	17	Button Head	1.312	.562
5306-00-240-0808	2.625	1.593	1.031	3/4-10	0.828	17	Button Head	1.312	.562
5306-00-240-0809	2.8125	1.031	1.031	3/4-10	0.828	17	Button Head	1.312	.562
5306-00-240-0810	2.625	1.406	1.219	7/8-9	0.953	15	Button Head	1.531	.562
5306-00-240-0811	3.1875	1.5312	1.219	7/8-9	0.953	15	Counter Sunk	1.584	.438
5306-00-240-0812	3.00	1.406	1.156	7/8-9	0.953	15	Counter Sunk	1.625	.438
5306-00-240-0813	2.500	0.906	1.156	7/8-9	0.953	15	Counter Sunk	1.625	.438

Dimensions are in inches

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Table II. Commercial Standard Dimensions for the ASTM - A325 Bolts Buttonhead Interference Body Type and ASTM F436 for Washers

Bolt Size
(The grip length includes allowance for one washer)
Dimensions are in inches

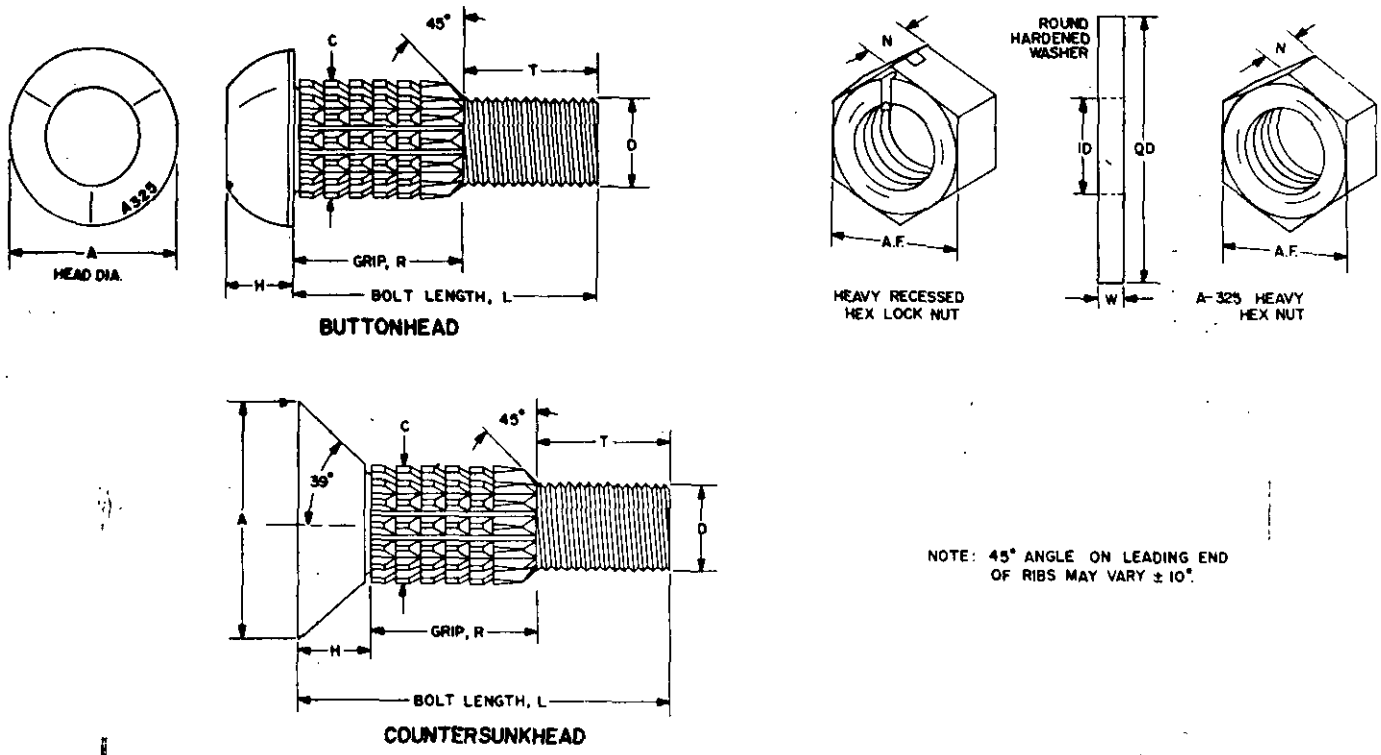
Bolt Diameter-D	5/8	3/4	7/8	1
Hole Size	11/16	13/16	15/16	1-1/16
Thickness of the Work (Grip)	Bolt Length L, in.	Bolt Length L, in.	Bolt Length L, in.	Bolt Length L, in.
5/16 to 3/8	1-3/16			
7/16 to 1/2	1-5/16	1-1/2		
9/16 to 5/8	1-7/16	1-5/8	1-3/4	
11/16 to 3/4	1-9/16	1-3/4	1-7/8	
13/16 to 7/8	1-11/16	1-7/8	2	2-1/8
15/16 to 1	1-13/16	2	2-1/8	2-1/4
1-1/16 to 1-1/8	1-15/16	2-1/8	2-1/4	2-3/8
1-3/16 to 1-1/4	2-1/16	2-1/4	2-3/8	2-1/2
1-5/16 to 1-3/8	2-3/16	2-3/8	2-1/2	2-5/8
1-7/16 to 1-1/2	2-5/16	2-1/2	2-5/8	2-3/4
1-9/16 to 1-5/8	2-7/16	2-5/8	2-3/4	2-7/8
1-11/16 to 1-3/4	2-9/16	2-3/4	2-7/8	3
1-13/16 to 1-7/8	2-11/16	2-7/8	3	3-1/8
1-15/16 to 2	2-13/16	3	3-1/8	3-1/4
2-1/16 to 2-1/8	2-15/16	3-1/8	3-1/4	3-3/8
2-3/16 to 2-1/4	3-1/16	3-1/4	3-3/8	3-1/2
2-5/16 to 2-3/8	3-3/16	3-3/8	3-1/2	3-5/8
2-7/16 to 2-1/2	3-5/16	3-1/2	3-5/8	3-3/4
2-9/16 to 2-5/8	3-7/16	3-5/8	3-3/4	3-7/8
2-11/16 to 2-3/4	3-9/16	3-3/4	3-7/8	4
2-13/16 to 2-7/8	3-11/16	3-7/8	4	4-1/8
2-15/16 to 3	3-13/16	4	4-1/8	4-1/4
3-1/16 to 3-1/8		4-1/8	4-1/4	4-3/8
3-3/16 to 3-1/4		4-1/4	4-3/8	4-1/2
3-5/16 to 3-3/8		4-3/8	4-1/2	4-5/8
3-7/16 to 3-1/2		4-1/2	4-5/8	4-3/4
3-9/16 to 3-5/8		4-5/8	4-3/4	4-7/8
3-11/16 to 3-3/4		4-3/4	4-7/8	5
3-13/16 to 3-7/8		4-7/8	5	5-1/8
3-15/16 to 4		5	5-1/8	5-1/4
4-1/16 to 4-1/8		5-1/8	5-1/4	5-3/8
4-3/16 to 4-1/4			5-3/8	5-1/2
4-5/16 to 4-3/8				5-5/8
4-7/16 to 4-1/2				5-3/4

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Table II. Commercial Standard Dimensions for the ASTM - A325 Bolts Buttonhead Interference Body Type and ASTM F436 for Washers (Cont'd)

Bolt Diameter-D	5/8	3/4	7/8	1
4-9/16 to 4-5/8				5-7/8
4-11/16 to 4-3/4				6
4-13/16 to 4-7/8				6-1/8
4-15/16 to 5				6-1/4
5-1/16 to 5-1/8				6-3/8
Head Diameter-A	1-9/64	1-3/8	1-19/32	1-13/16
Head Height-H	25/64	15-32	35/64	39/64
Diameter Across Ribs-C	45/64	53/64	61/64	1-5/64
Thread Length-T	27/32	1-1/32	1-5/32	1-9/32
Nut Thickness-N, in.	5/8	3/4	7/8	1
Nut, Across Flats-A.F.	1-1/16	1-1/4	1-7/16	1-5/8
Washer O.D.	1-1/2	1-3/4	2	2-1/4
Washer I.D.	11/16	13/16	15/16	1-1/16
Washer Thickness (nom)-W	0.134	0.148	0.165	0.165

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NOTE: 45° ANGLE ON LEADING END OF RIBS MAY VARY ± 10°.

FIGURE 1. Bolt, ribbed neck, interference body.

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3.4 Workmanship. The bolts and nuts, and washers when furnished, shall be so designed and fabricated as to insure a body bound fit and a permanent fastening when installed. All threading and mating surfaces shall be free from burrs, and other surface irregularities.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major defects specified in 4.4.1. The AQL shall be 4.0 percent defective for major defects.

4.4 Inspection procedure.

4.4.1 Examination. The bolts and nuts shall be examined as specified herein for the following defects:

Major

101. Bolts and nuts not as shown in Table I and Figure 1.
102. Material not as specified.
103. Threads not as specified.
104. Boltheads not as specified.
105. Size not as specified.
106. Workmanship not as specified.

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4.5 Inspection of packaging. Examination of the preservation, packing and marking of the bolts and nuts shall be in accordance with the Quality Assurance Provisions of PPP-H-1581.

5. PACKAGING

5.1 Preservation, packing, and marking. The bolts and nuts shall be preserved, packed, and marked in accordance with PPP-H-1581. Level A preservation, Level A and B packing, and military packaging marking shall be in accordance with PPP-H-1581 for the level specified. Commercial preservation and commercial packing shall be in accordance with the Level C requirements of PPP-H-1581. Commercial packaging marking shall reflect as a minimum, the item name, National Stock Number, quantity, contract number, and appropriate address markings. The marking shall be applied with a waterproof material, be legible and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The interrupted rib, interference type bolts with nuts are intended for use in bridge construction in lieu of rivets. This revision of MIL-B-52796(ME) is intended to replace CE PD-236, dated 15 September 1952. Any reference to PD-236 for further procurements should be referred to this revision.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type bolthead required (see 3.3.2, Table I, and Figure 1).
(c) Size of bolts required (see 3.3.3, and Tables I and II).
(d) When washers are required (see 3.1).
(e) When commercial bolts are required (see 3.4).
(f) Degree of preservation and degree of packing required (see 5.1).

6.3 Bolt sizes. The bolt sizes and types covered in Table I are intended for use with Bridge Equipment Sets SC 5420-97-CL-E10, 5420-97-CL-E11, and 5420-97-CL-E12.

6.4 Bolt characteristics .

6.4.1 Bolts used with washers. Button-head or countersunk-head bolts when used with washers should be used with heavy hex (nonrecessed) nuts of the non-locking type. The washers prevent contact between the bolt ribs and the nut while the high clamping loads minimize the need for locknuts.

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6.4.2 Bolts without washers. Button-head or countersunk-head bolts when used with recessed locknuts of the type illustrated in Figure 1 should be used without washers. The recess in the nut provides safety against the contact between bolt ribs and the nut. The distance from the work face to the crown of the nut provides adequate "stick through" of the bolt to assure good contact between the lock wire and the bolt threads thereby providing resistance to slip of the clamped members and the locknut assures a secure connection.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-B-52796A(ME) Bolts, Ribbed Neck, Structural Type, with Recessed Locknut

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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